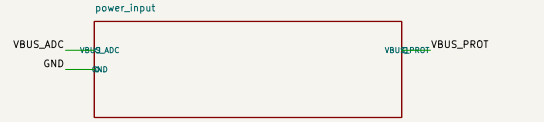
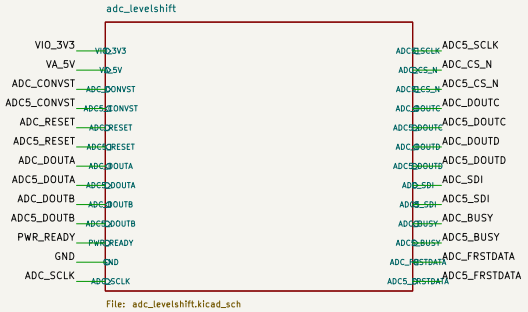
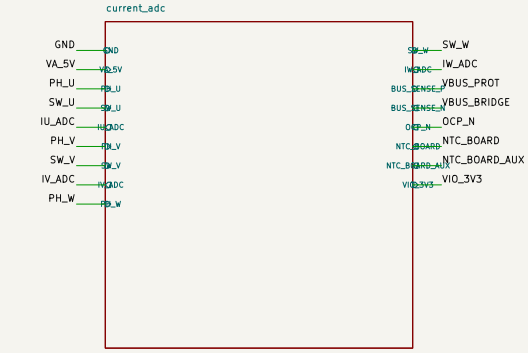
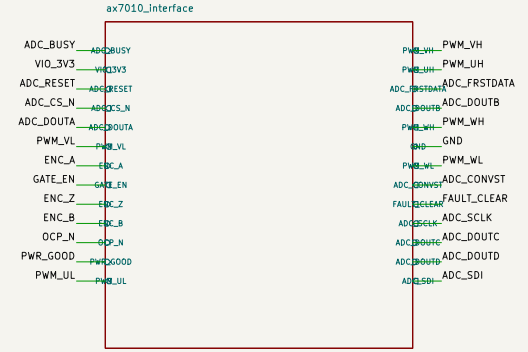
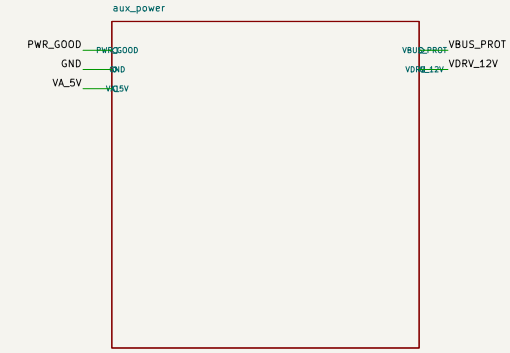
INDEPENDENT AD7606C ENGINEERING VARIANT — ORIGINAL REV.A RETAINED

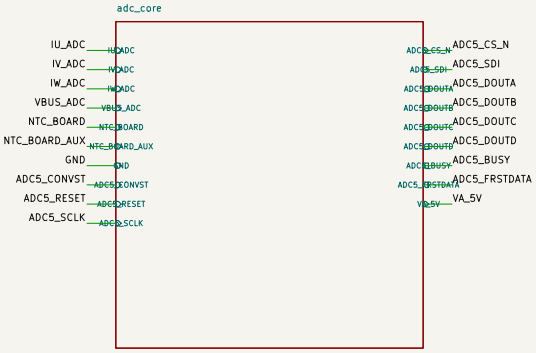


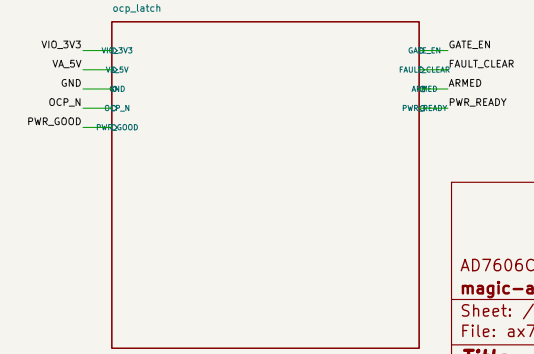



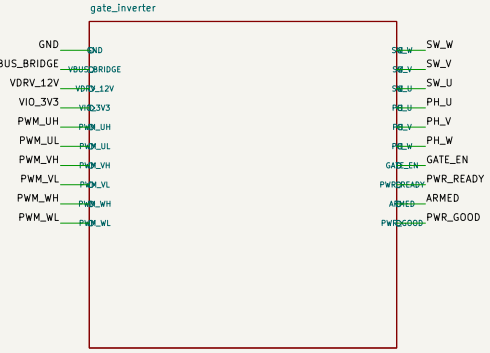



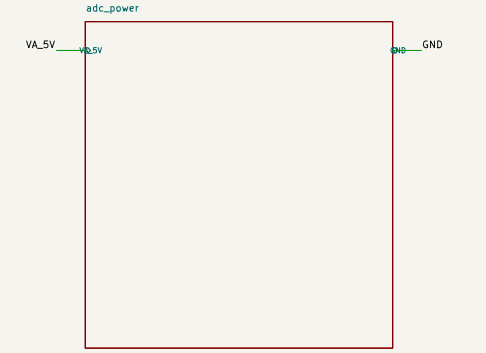







AD7606C independent development schematic; release gates OPEN

**magic-alt/fpga-servo**

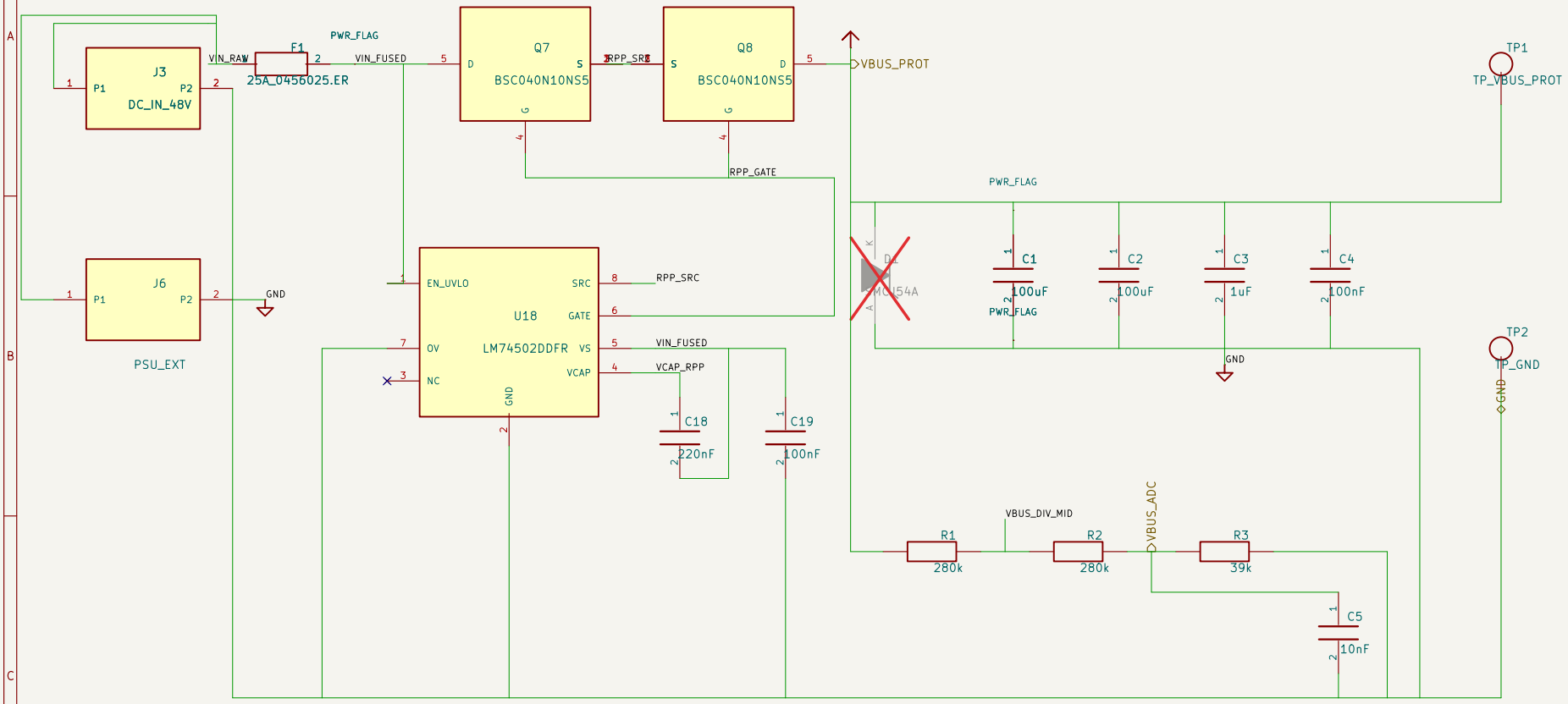
Sheet: /

File: ax7010_servo_ad7606c.kicad_sch

**Title: AX7010 servo — AD7606C Rev.B**

| | | |
|---|---|---|
| Size: A4 | Date: 2026-10-17 | **Rev: B-DEV** |
| KiCad E.D.A. 10.0.3 | | Id: 1/11 |

SOURCE + -> J3.1 -> ONBOARD F1 -> LM74502 RPP -> DC-LINK




Q7/Q8 are back-to-back N-MOSFETs driven by LM74502 for low-loss reverse-polarity protection. Reverse current is not blocked when enabled; regeneration needs a receptive source.

LM74502 OV is tied low in Rev.A1; positive surge is NOT claimed as closed. The 48V source/cable transient must be measured and kept inside the controller/downstream qualification envelope.

F1: ONBOARD 0456025.ER 25A SMD. Verify fault current, inrush and thermal coordination. D1 stays DNP pending surge qualification.

VBUS_ADC divider = 280k + 280k over 39k, plus 10nF at the ADC node. Keep the high-side resistors split for voltage stress and route the sense return away from commutation current.

Local wiring: power, gate, divider/filter and ground. See docs/schematic_full_review_2026-10-09.md.

24..48V operating bus; source transient / regeneration qualification OPEN

**magic-alt/fpga-servo**

Sheet: /power_input/

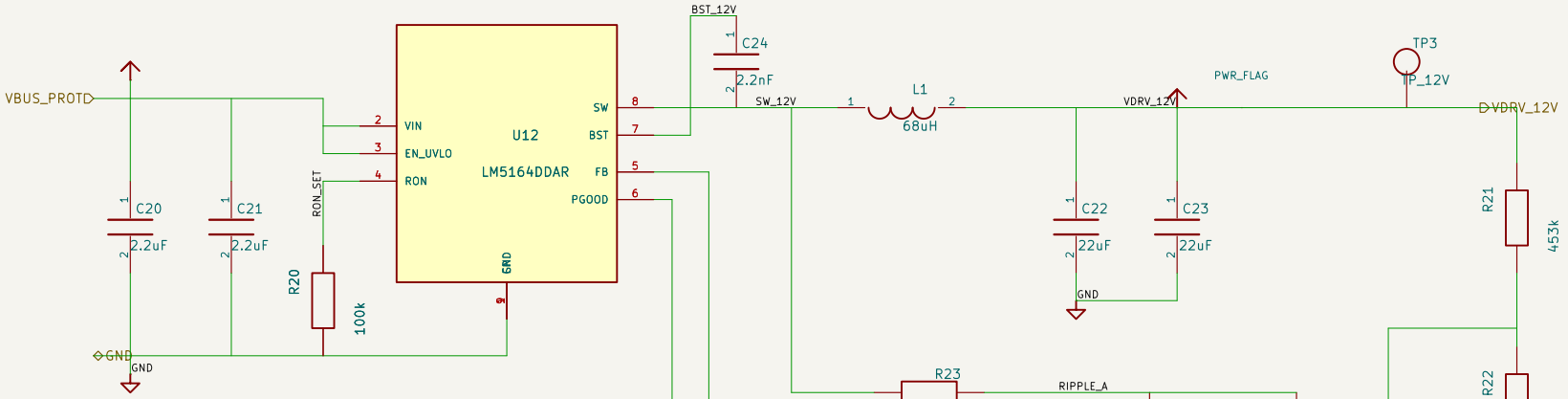
File: power_input.kicad_sch

***Title: Rev.A1 — DC Input & Protection***

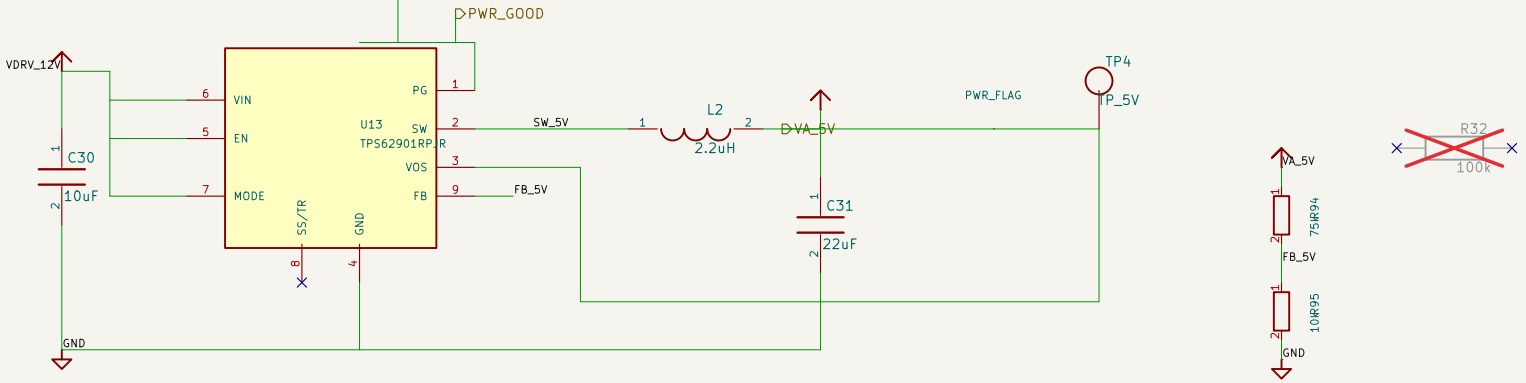
| Size: A4 | Date: 2026-09-29 | **Rev: A1** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 2/11 |

**VBUS_PROT -> LM5164 12V -> TPS62901 5.1V. Rail load / transient qualification OPEN.**

12 V BUCK: input / bootstrap / output LC / feedback + ripple injection




5 V BUCK: precision external FB; forced PWM 2.5MHz




LM5164 values follow TI Figure 7-1: 15..100V in, 12V/1A, 300kHz, 68uH, 2x2.2uF CIN, 2x22uF COUT, 453k/49.9k feedback.

TPS62901: 5.1V nominal; FB=75k/10k; MODE=VIN: forced PWM 2.5MHz. VOS to output.

LM5164 PGOOD and TPS62901 PG share PWR_GOOD; pull up at 3.3V safety logic.

System operating input: 24..48V. Low-voltage bring-up is separately qualified.

PGOOD alone does not prove VA5 valid; U25 separately monitors VA5.

LM5164 48V->12V + TPS62901 precision 5.1V

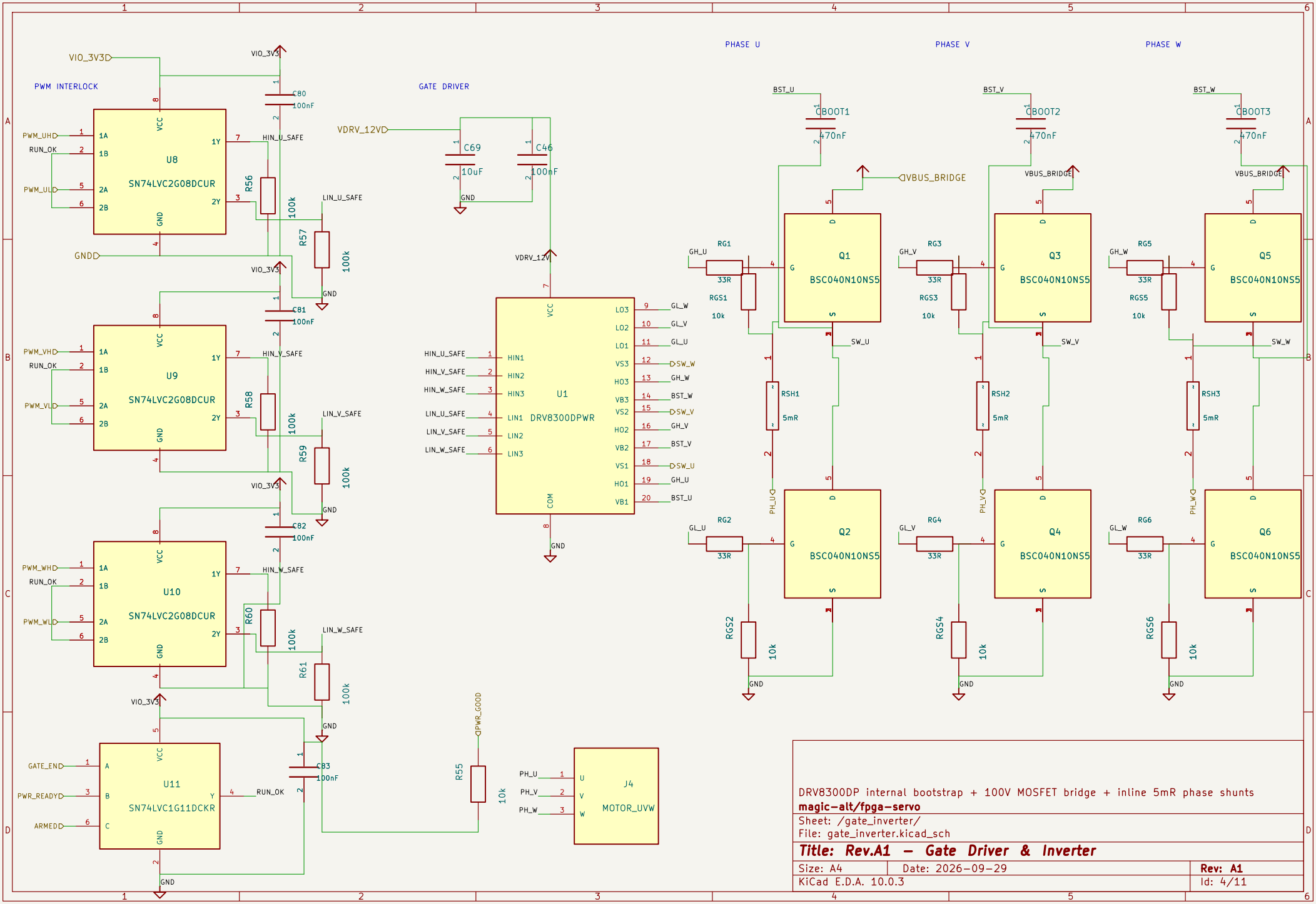
**magic-alt/fpga-servo**

Sheet: /aux_power/

File: aux_power.kicad_sch

***Title: Rev.A1 - Auxiliary Power***

| | | |
|---|---|---|
| Size: A4 | Date: 2026-09-29 | **Rev: A1** |
| KiCad E.D.A. 10.0.3 | | Id: 3/11 |


PWM INTERLOCK
GATE DRIVER
PHASE U
PHASE V
PHASE W
VIO_3V3
VDRV_12VD
VDRV_12V
GNDD
GND
VBUS_BRIDGE
U8
U9
U10
U11
U1
SN74LVC2G08DCUR
SN74LVC1G11DCKR
DRV8300DPWR
PWM_UH
PWM_UL
PWM_VH
PWM_VL
PWM_WH
PWM_WL
RUN_OK
GATE_EN
PWR_READY
ARMED
HIN_U_SAFE
HIN_V_SAFE
HIN_W_SAFE
LIN_U_SAFE
LIN_V_SAFE
LIN_W_SAFE
C80
C81
C82
C83
C69
C46
CBOOT1
CBOOT2
CBOOT3
100nF
10uF
470nF
R56
R57
R58
R59
R60
R61
R55
100k
10k
RG1
RG2
RG3
RG4
RG5
RG6
RGS1
RGS2
RGS3
RGS4
RGS5
RGS6
33R
RSH1
RSH2
RSH3
5mR
Q1
Q2
Q3
Q4
Q5
Q6
BSC040N10NS5
BST_U
BST_V
BST_W
SW_U
SW_V
SW_W
GH_U
GH_V
GH_W
GL_U
GL_V
GL_W
PH_U
PH_V
PH_W
PWR_GOOD
J4
MOTOR_UVW
DRV8300DP internal bootstrap + 100V MOSFET bridge + inline 5mR phase shunts
magic-alt/fpga-servo
Sheet: /gate_inverter/
File: gate_inverter.kicad_sch
Title: Rev.A1 – Gate Driver & Inverter
Size: A4
Date: 2026-09-29
Rev: A1
KiCad E.D.A. 10.0.3
Id: 4/11

Phase RC: 47R / 1nF retained for first prototype; tune after PWM-noise tests

BUS OCP: +25A NOMINAL

CURRENT SENSE + RC FILTER

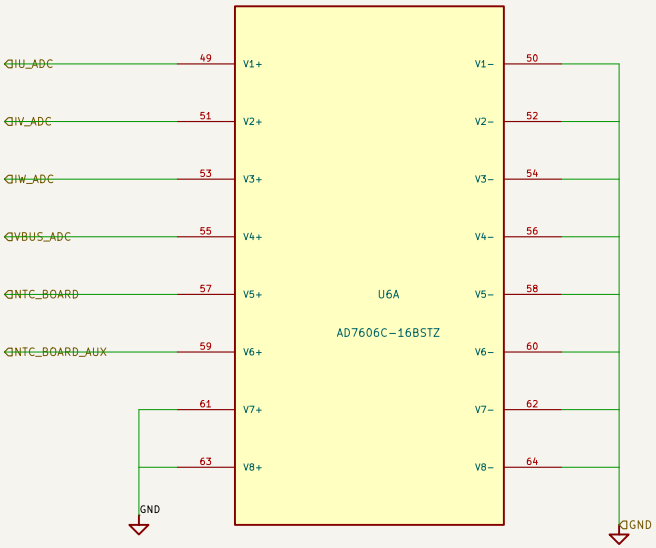
Bridge bus shunt after DC-link capacitors

ADC moved to dedicated core + power sheets

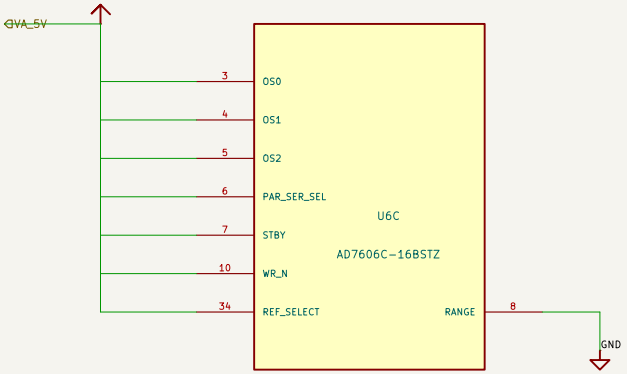
No reverse-current trip; latch requires explicit re-arm

| | | |
|---|---|---|
| AD7606C independent development schematic; release gates OPEN | | |
| **magic-alt/fpga-servo** | | |
| Sheet: /current_adc/ | | |
| File: current_adc.kicad_sch | | |
| ***Title: Rev.B – current sensing / independent bus OCP*** | | |
| Size: A4 | Date: 2026-10-17 | **Rev: B-DEV** |
| KiCad E.D.A. 10.0.3 | | Id: 5/11 |

AD7606C-16 / ANALOG INPUTS




MODE STRAPS: software / serial / internal reference

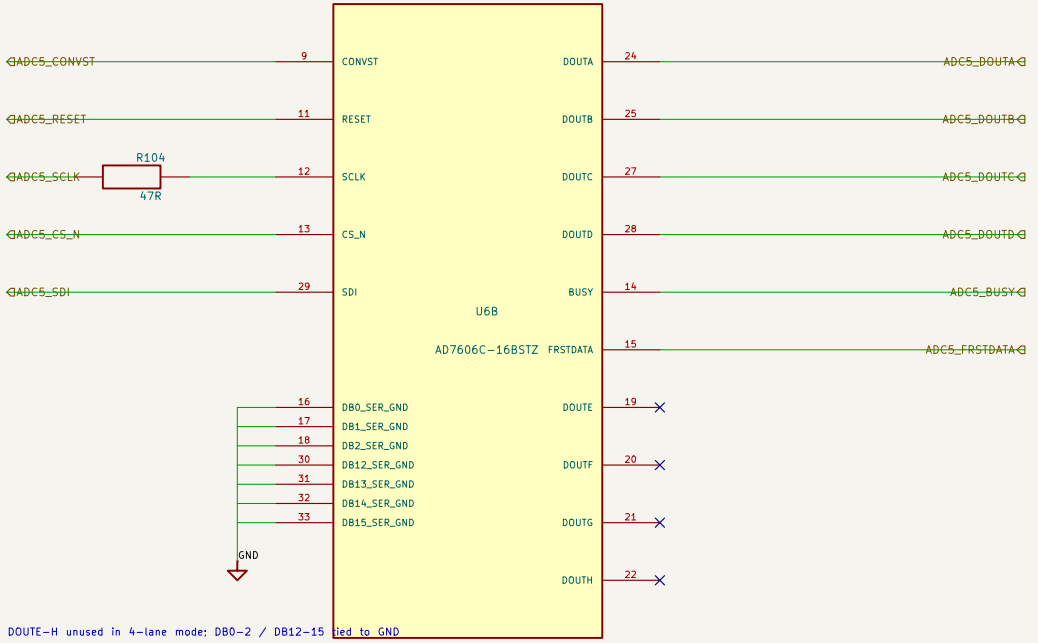



Hold GATE_EN low until reset + configuration readback pass.

Unipolar ADC codes: straight binary. See adc_profile.json.

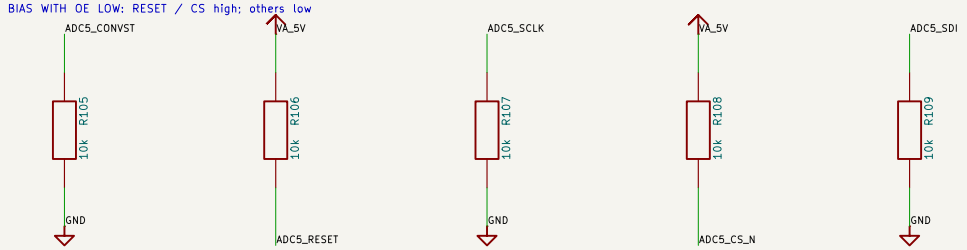
Register writes are required after power-up; straps alone do not configure ranges.

4-LANE / 5 MHz BRING-UP / 20 MHz QUALIFICATION TARGET




DOUTE-H unused in 4-lane mode; DB0-2 / DB12-15 tied to GND

BIAS WITH OE LOW: RESET / CS high; others low




AD7606C independent development schematic; release gates OPEN

**magic-alt/fpga-servo**

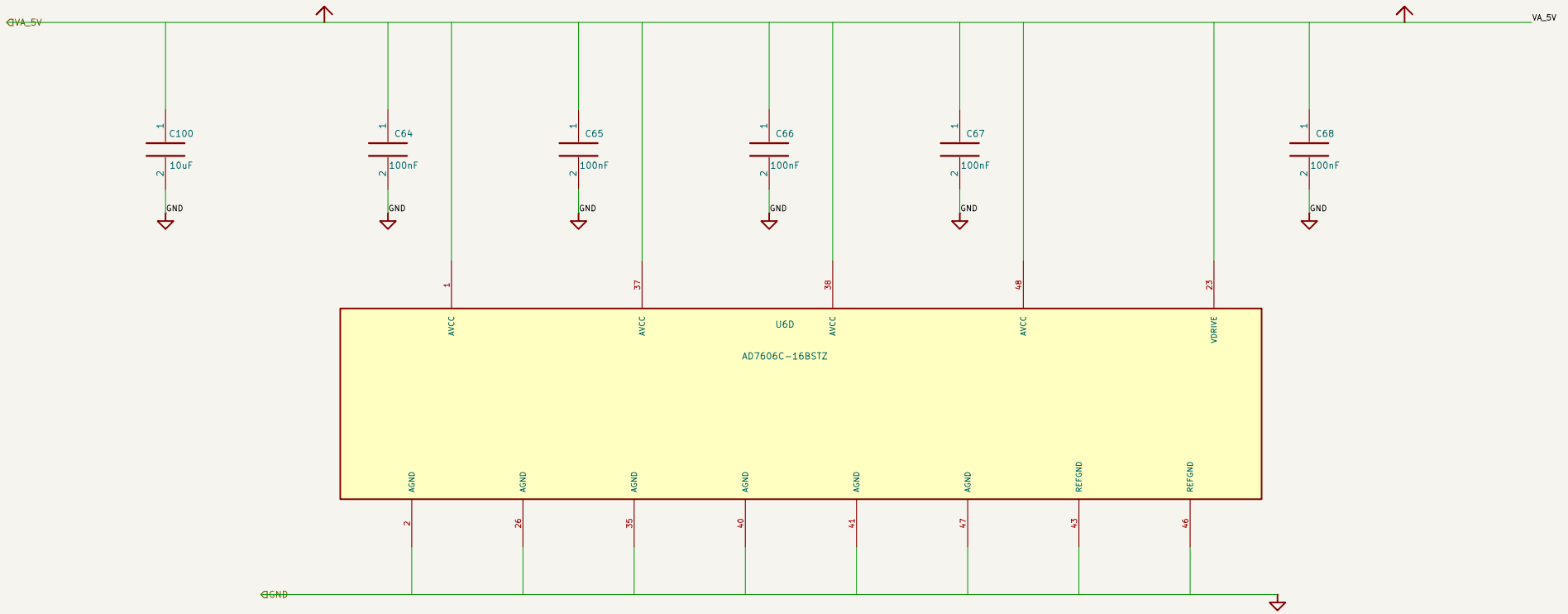
Sheet: /adc_core/

File: adc_core.kicad_sch

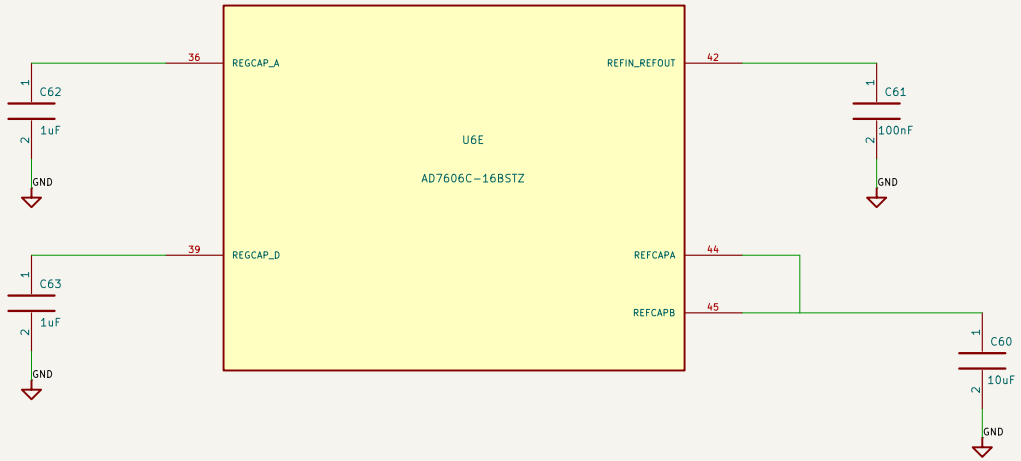
***Title: Rev.B - AD7606C input / serial / startup***

| Size: A4 | Date: 2026-10-17 | **Rev: B-DEV** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 6/11 |

ADC SUPPLIES: 100 nF AT EACH AVCC + 10 uF LOCAL BULK




REGCAP A/D SEPARATE; REFCAPA/B SHORTED; NO EXTERNAL LOAD ON THESE NODES




AD7606C independent development schematic; release gates OPEN

**magic-alt/fpga-servo**

Sheet: /adc_power/

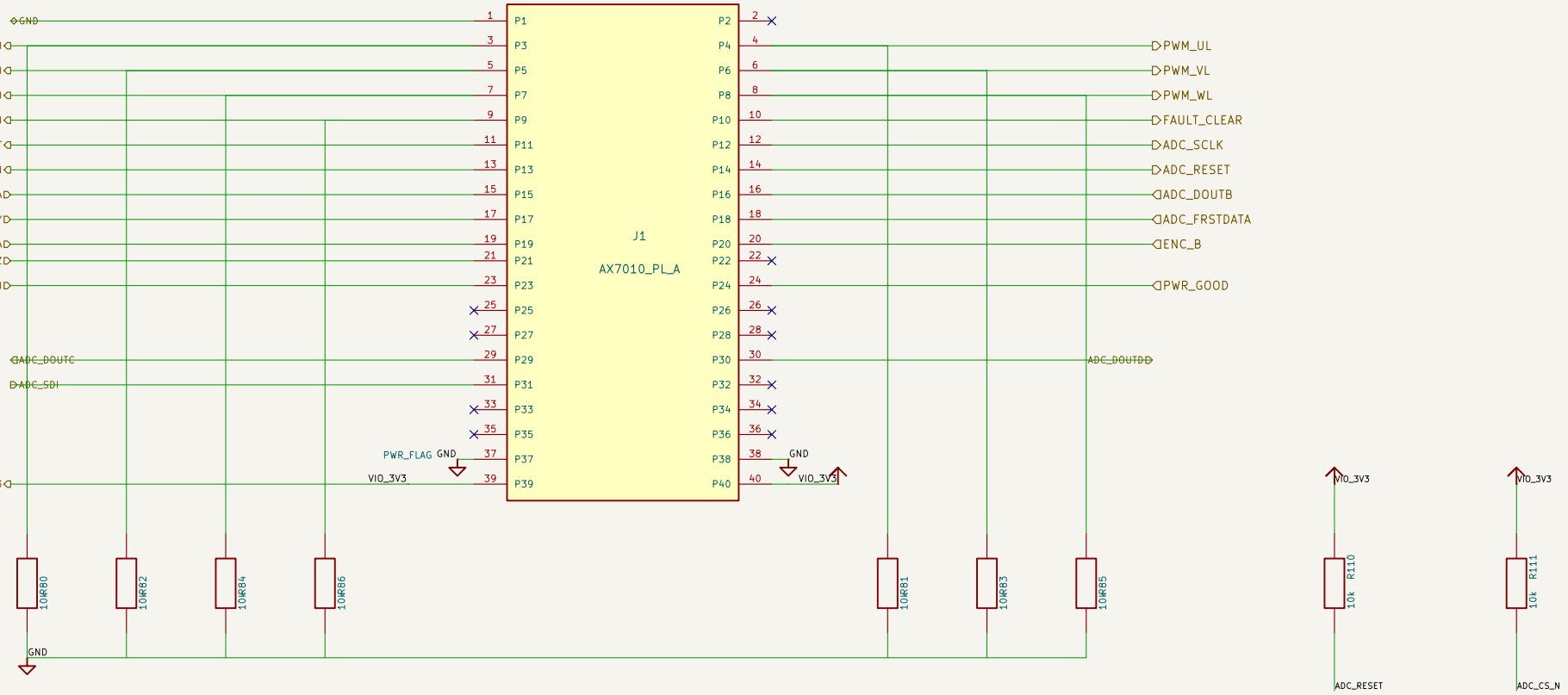
File: adc_power.kicad_sch

***Title: Rev.B — AD7606C supply / reference / LDO bypass***

| | | |
|---|---|---|
| Size: A4 | Date: 2026-10-17 | **Rev: B-DEV** |
| KiCad E.D.A. 10.0.3 | | Id: 7/11 |

ALINX AX7010 (2022): active J10 interface + physical reset-state pull-down branches

J1 = AX7010 J10




R80..R86: each PWM and GATE_EN line has its own 10k pull-down.

J10 pin 2 (+5 V): NC. J2 removed; only J1 interfaces to AX7010.

AD7606C: J10.29=DOUTC (U12), .30=DOUTD (T12), .31=SDI (T10)

R110/R111: hold RESET/CS high at TXU inputs while FPGA is Hi-Z.

AD7606C independent development schematic; release gates OPEN

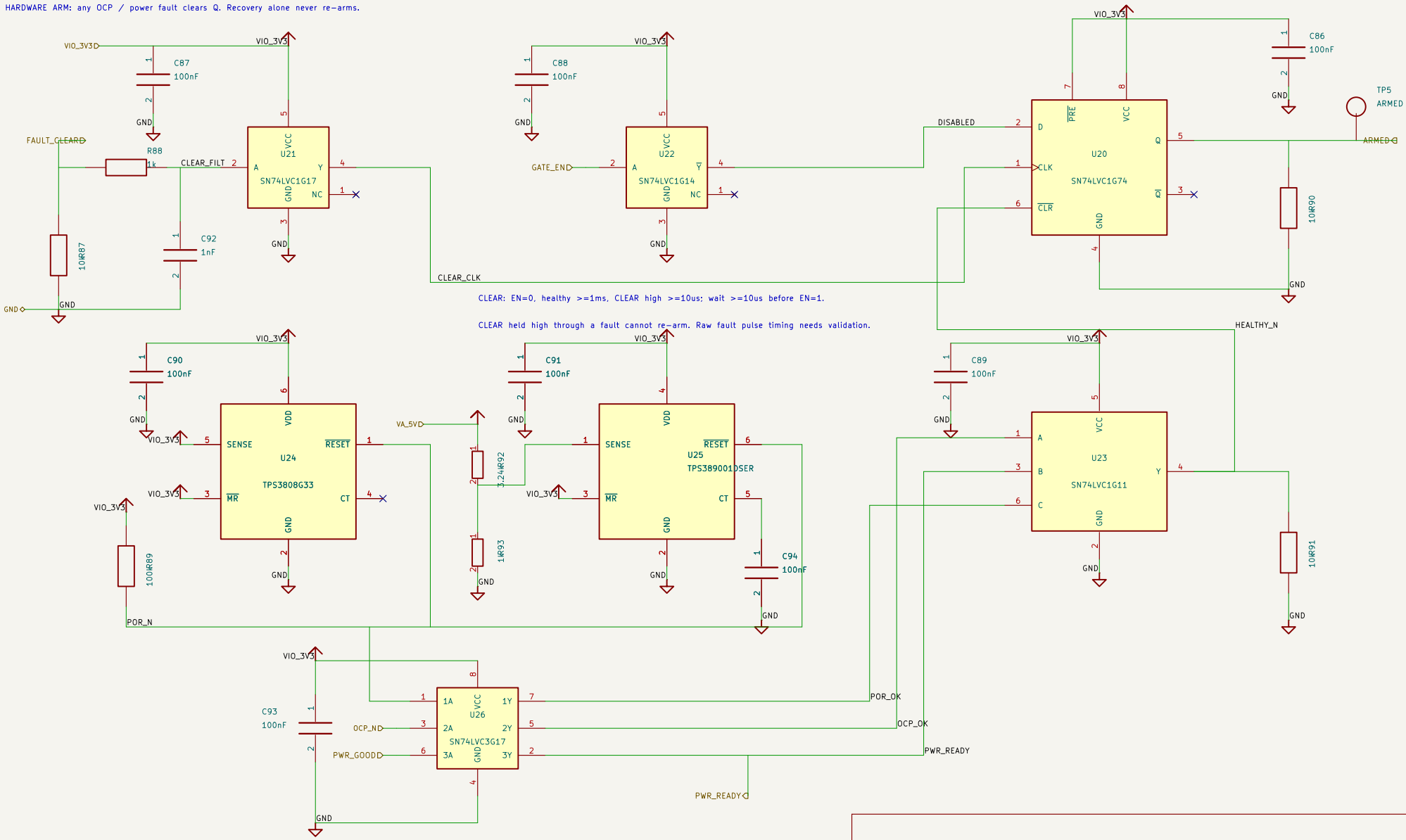
**magic-alt/fpga-servo**

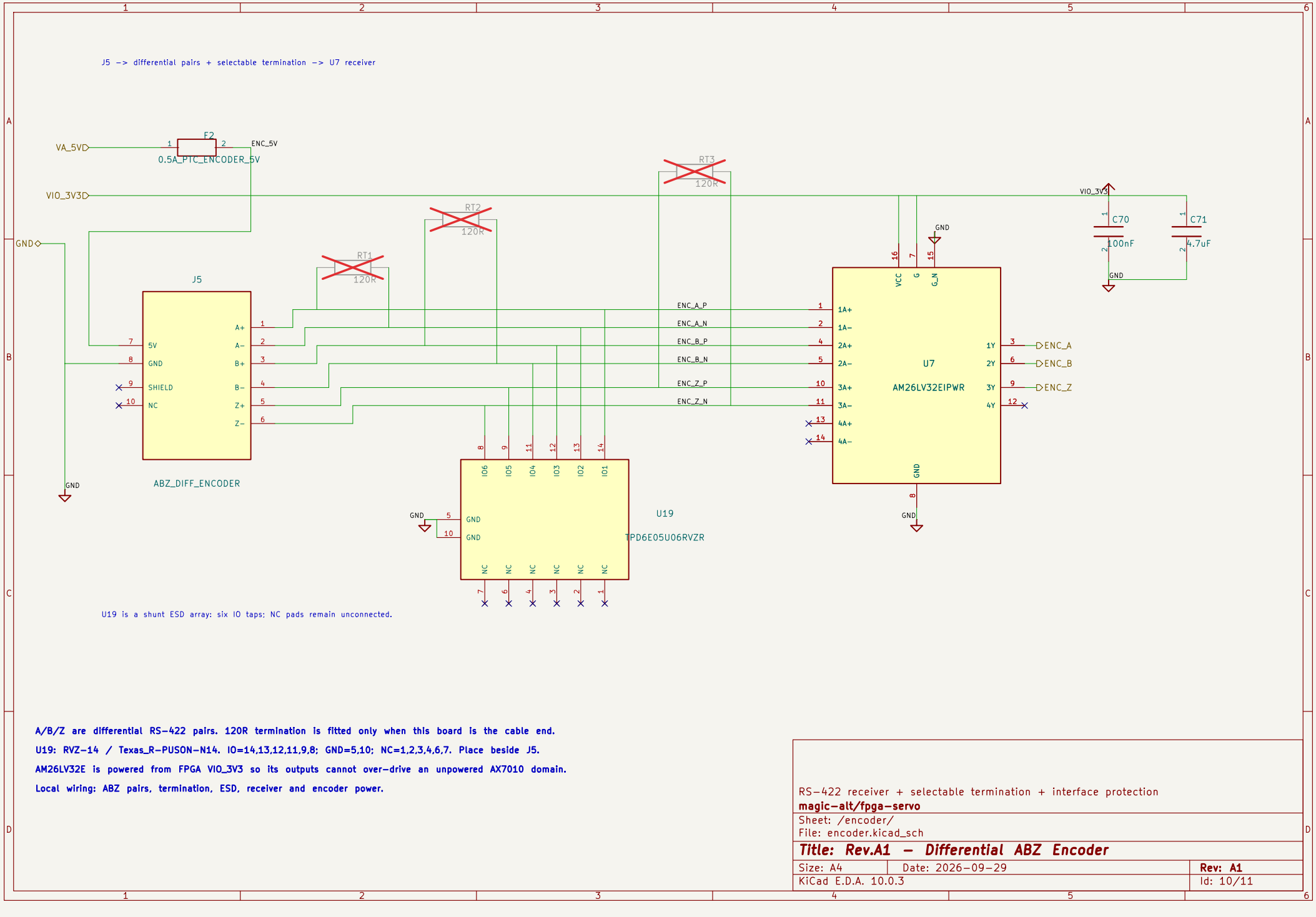
Sheet: /ax7010_interface/

File: ax7010_interface.kicad_sch

***Title: Rev.B – AX7010 2022 J10 / four-lane ADC***

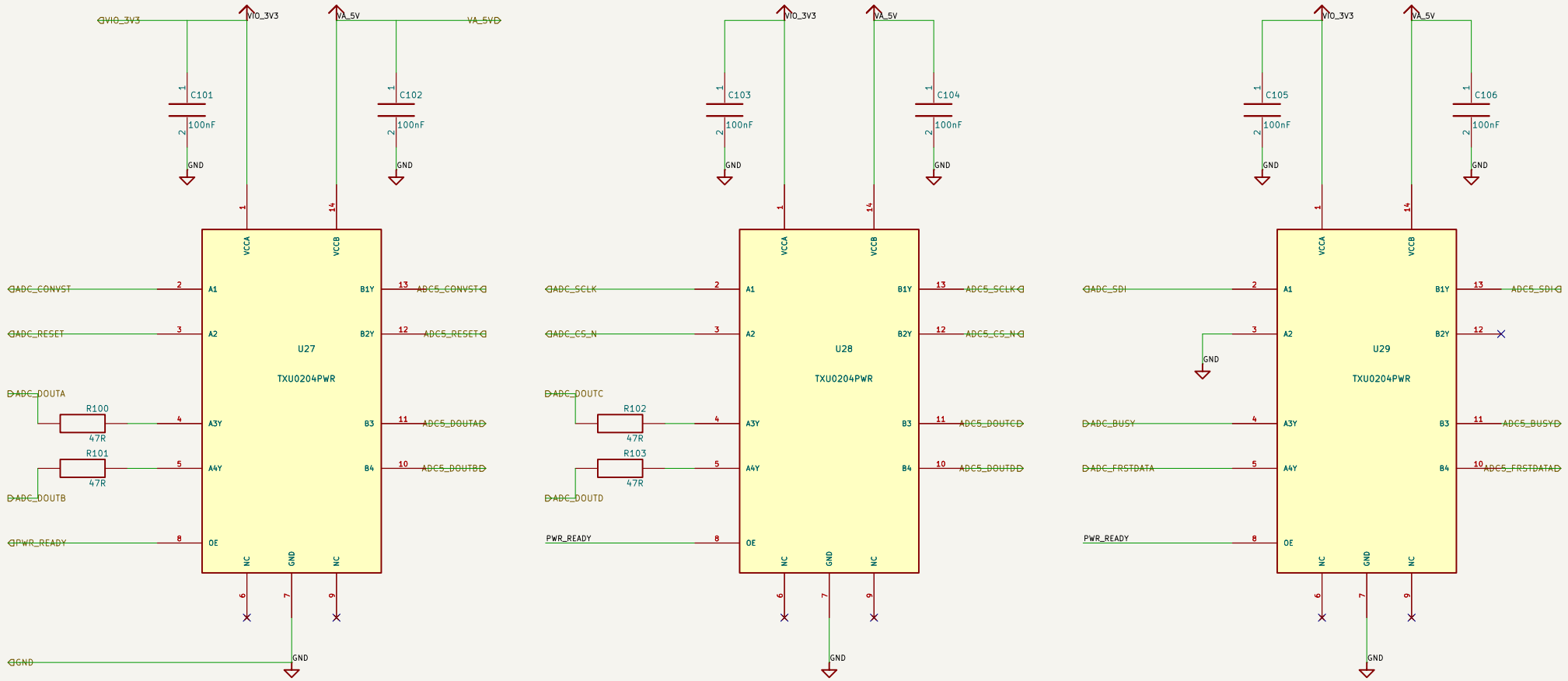
| Size: A4 | Date: 2026-10-17 | **Rev: B-DEV** |
|---|---|---|
| KiCad E.D.A. 10.0.3 | | Id: 8/11 |


HARDWARE ARM: any OCP / power fault clears Q. Recovery alone never re-arms.
VIO_3V3
C87
100nF
GND
FAULT_CLEARD
R88
1k
CLEAR_FILT
U21
SN74LVC1G17
VCC
A
Y
GND
NC
C92
1nF
10kR87
C88
100nF
GATE_EN
U22
SN74LVC1G14
DISABLED
U20
SN74LVC1G74
D
CLK
PRE
Q
CLR
C86
100nF
TP5
ARMED
10kR90
CLEAR_CLK
CLEAR: EN=0, healthy >=1ms, CLEAR high >=10us; wait >=10us before EN=1.
CLEAR held high through a fault cannot re-arm. Raw fault pulse timing needs validation.
HEALTHY_N
C90
100nF
U24
TPS3808G33
SENSE
RESET
MR
CT
VDD
VA_5V
3.24kR92
1kR93
C91
100nF
U25
TPS3890010SER
C89
100nF
U23
SN74LVC1G11
10kR91
100kR89
POR_N
C94
100nF
C93
100nF
U26
SN74LVC3G17
1A
2A
3A
1Y
2Y
3Y
OCP_ND
PWR_GOODD
POR_OK
OCP_OK
PWR_READY
U24 CT open: 12..28ms. U25 CT=100nF: ~107ms; VA5 trip nominal 4.876V. Dynamic qualification OPEN.
U20..U26: local 100nF bypass; reset release passes through U26 Schmitt buffers.
magic-alt/fpga-servo
Sheet: /ocp_latch/
File: ocp_latch.kicad_sch
Title: Hardware OCP latch / supervised re-arm
Size: A4
Date: 2026-10-09
Rev: A1
KiCad E.D.A. 10.0.3
Id: 9/11


J5 -> differential pairs + selectable termination -> U7 receiver
VA_5VD
F2
0.5A_PTC_ENCODER_5V
ENC_5V
VIO_3V3D
GND
RT1
120R
RT2
120R
RT3
120R
J5
A+
A-
B+
B-
Z+
Z-
5V
GND
SHIELD
NC
ABZ_DIFF_ENCODER
ENC_A_P
ENC_A_N
ENC_B_P
ENC_B_N
ENC_Z_P
ENC_Z_N
U7
AM26LV32EIPWR
VCC
G
G_N
1A+
1A-
2A+
2A-
3A+
3A-
4A+
4A-
1Y
2Y
3Y
4Y
ENC_A
ENC_B
ENC_Z
VIO_3V3
C70
100nF
C71
4.7uF
U19
TPD6E05U06RVZR
IO6
IO5
IO4
IO3
IO2
IO1
NC
U19 is a shunt ESD array: six IO taps; NC pads remain unconnected.
A/B/Z are differential RS-422 pairs. 120R termination is fitted only when this board is the cable end.
U19: RVZ-14 / Texas_R-PUSON-N14. IO=14,13,12,11,9,8; GND=5,10; NC=1,2,3,4,6,7. Place beside J5.
AM26LV32E is powered from FPGA VIO_3V3 so its outputs cannot over-drive an unpowered AX7010 domain.
Local wiring: ABZ pairs, termination, ESD, receiver and encoder power.
RS-422 receiver + selectable termination + interface protection
magic-alt/fpga-servo
Sheet: /encoder/
File: encoder.kicad_sch
Title: Rev.A1 - Differential ABZ Encoder
Size: A4
Date: 2026-09-29
Rev: A1
KiCad E.D.A. 10.0.3
Id: 10/11

ADC POWER-DOMAIN ISOLATION: FPGA 3.3 V <-> ADC VA_5V

U6 AVCC = VDRIVE = VA_5V; never connect AX7010 3.3 V directly to U6.23




OE follows PWR_READY: translators disable on supervised rail faults.

TI Ioff / VCC disconnect prevents back-power when either supply is off.

RESET / CS pulls are on the ADC side. Firmware must apply full RESET after OE is enabled.

Translate delay, J10 flight time and ADC delay must all be included in Vivado I/O timing.

AD7606C independent development schematic; release gates OPEN

**magic-alt/fpga-servo**

Sheet: /adc_levelshift/

File: adc_levelshift.kicad_sch

***Title: Rev.B – ADC voltage / partial-power isolation***

| | | |
|---|---|---|
| Size: A4 | Date: 2026-10-17 | **Rev: B-DEV** |
| KiCad E.D.A. 10.0.3 | | Id: 11/11 |